

Conformal Prediction Under Distribution Shift: A COVID-19 Natural Experiment

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Abstract

We study how conformal prediction guarantees degrade under distribution shift using COVID-19 as a natural experiment. Analyzing 8 supply chain tasks experiencing severe feature turnover (Jaccard similarity ≈ 0), we find coverage drops ranging from 0% to 86.7%—two orders of magnitude variation under identical temporal shift. **Predictive signal:** Using SHAP analysis, we identify that catastrophic failures correlate with *single-feature dependence* rather than feature instability (Spearman $\rho = 0.714$, $p = 0.047$). Both catastrophic and robust tasks exhibit $\sim 0\%$ feature overlap (Jaccard similarity), yet coverage drops differ by $700\times$ (71.6% vs 0.1%) due to feature importance dynamics: catastrophic tasks show concentrated importance explosion in one feature ($4.5\times$), while robust tasks redistribute importance across multiple features ($10\text{--}20\times$). **Solution:** Retraining experiments over 11 months show quarterly retraining significantly restores catastrophic task coverage from 22% to 41% (+19 percentage points, Wilcoxon $p = 0.04$ vs no retraining). While quarterly achieves higher mean coverage than monthly retraining (41% vs 32%), this difference is not statistically significant ($p = 0.24$); quarterly’s advantage lies in cost-effectiveness (3 vs 10 retrains/year) and stability. Robust tasks maintain 99.8% coverage regardless of retraining, confirming retraining is unnecessary for tasks with distributed importance. **Exploratory validation:** Analysis of 4 additional tasks from clinical trials and motorsports with moderate feature stability (Jaccard 0.13–0.86) reveals a different mechanism—feature stability (not concentration) determines robustness, suggesting the concentration effect applies specifically to severe-shift scenarios. We provide a decision framework: monitor SHAP importance concentration before deployment; for vulnerable tasks ($>40\%$ concentration), implement quarterly retraining; for robust tasks, skip retraining to save cost.

CCS Concepts

• **Computing methodologies** $\rightarrow$ **Uncertainty quantification**; • **Mathematics of computing** $\rightarrow$ *Time series analysis*.

Keywords

Conformal Prediction, Distribution Shift, COVID-19, Uncertainty Quantification, Supply Chain

1 Introduction

Conformal prediction provides distribution-free coverage guarantees under the assumption of exchangeability [14]. However,

real-world deployments face distribution shifts that violate this assumption. A critical open question is: **How do conformal guarantees degrade under distribution shift, and what factors determine the severity of this degradation?**

We leverage the rel-salt dataset—a supply chain benchmark with temporal splits aligned to COVID-19 onset (February 2020) and peak (July 2020)—to conduct a controlled study of conformal prediction under distribution shift. Our key contributions:

- (1) **Quantification:** We measure coverage degradation across 8 supply chain tasks experiencing severe feature turnover (Jaccard ≈ 0), finding drops ranging from 0% to 86.7%—two orders of magnitude variation under identical temporal shift
- (2) **Predictive Signal:** Using SHAP analysis, we identify single-feature dependence as a predictive signal for catastrophic failure in severe-shift scenarios (Spearman $\rho = 0.714$, $p = 0.047$, $n=8$). Exploratory analysis of 4 additional tasks with moderate feature stability (Jaccard 0.13–0.86) reveals a different mechanism, suggesting concentration effects apply specifically when features lack temporal stability
- (3) **Practical Solution:** Retraining experiments show quarterly retraining significantly restores catastrophic task coverage by 19 percentage points (Wilcoxon $p = 0.04$ vs no retraining). While quarterly achieves higher mean coverage than monthly (41% vs 32%), this difference is not statistically significant ($p = 0.24$)
- (4) **Cost-Benefit Analysis:** Robust tasks (99.8% coverage) gain no benefit from retraining, enabling practitioners to avoid unnecessary computational cost
- (5) **Decision Framework:** We provide actionable guidance based on SHAP importance concentration: monitor before deployment, retrain quarterly if vulnerable ($>40\%$ concentration), skip retraining if robust
- (6) **Negative Result:** We show that Adaptive Conformal Inference does not help under severe distribution shift

2 Related Work

Conformal Prediction. Vovk et al. [14] introduced conformal prediction with exchangeability guarantees. Recent work extends to classification [12] and regression [11].

Distribution Shift. Tibshirani et al. [13] study conformal prediction under covariate shift. We focus on temporal shift with feature staleness—a distinct failure mode.

Adaptive Methods. Gibbs and Candès [4] propose Adaptive Conformal Inference (ACI) for non-stationary settings. We test whether ACI helps under severe distribution shift.

2.1 Distribution Shift Benchmarks

Recent benchmarks systematically study distribution shift across domains. WILDS [6] provides standardized evaluation for in-the-wild shifts, while Shifts [9] focuses on safety-critical applications. Gulrajani and Lopez-Paz [5] show that many “robust” methods fail under temporal shift. We focus specifically on how *conformal prediction guarantees* degrade under temporal shift.

2.2 Conformal Prediction Under Distribution Shift

Tibshirani et al. [13] study conformal prediction under covariate shift with known propensity scores. Podkopaev and Ramdas [10] develop distribution-free prediction sets under label shift. Barber et al. [2] provide a comprehensive treatment of conformal prediction beyond exchangeability. We focus on *temporal shift with feature staleness*—a distinct failure mode where feature distributions have zero overlap between train and test.

Gibbs and Candès [4] propose Adaptive Conformal Inference (ACI) for non-stationary settings, with extensions by Zaffran et al. [15] for time series. We show in Section 5.1 that these adaptive methods *fail* when feature overlap is $\sim 0\%$ —this is not a calibration problem but a fundamental data shift that requires retraining.

2.3 Interpretability and Model Debugging

Our finding that SHAP importance concentration predicts conformal failure connects to work on feature attribution [8] and model debugging [1]. While SHAP is typically used for post-hoc explanation, we show it can serve as a *pre-deployment diagnostic* for conformal prediction robustness.

3 Methodology

3.1 Dataset: rel-salt

We use the rel-salt supply chain dataset with temporal splits:

- **Train:** Before February 2020 (pre-COVID)
- **Validation:** February–July 2020 (COVID onset)
- **Test:** After July 2020 (COVID peak)

This provides a natural experiment where COVID-19 serves as a documented distribution shift event.

3.2 Conformal Prediction Setup

We use conformal prediction via the Adaptive Prediction Sets (APS) algorithm [12] with $\alpha = 0.1$ (90% target coverage, computed as $(1 - \alpha) \times 100\%$):

- (1) Train LightGBM ensemble (50 seeds) on training data
- (2) Calibrate conformal predictor on 50% of validation set
- (3) Evaluate coverage on held-out validation and test sets

3.3 Ensemble for Variance Quantification

We train 50 independent models with random seeds 42–91 to quantify uncertainty due to model initialization. For each seed s :

- (1) Train a separate LightGBM classifier M_s with random seed s
- (2) Split the validation set 50/50 into calibration ($\mathcal{D}_{\text{cal}}^s$) and evaluation ($\mathcal{D}_{\text{eval}}^s$) sets
- (3) Calibrate an individual conformal predictor CP_s on $\mathcal{D}_{\text{cal}}^s$
- (4) Evaluate coverage on $\mathcal{D}_{\text{eval}}^s$ (validation coverage) and test set (test coverage)

We report mean $\pm$ standard deviation across 50 trials. This is *not* ensemble prediction (averaging model outputs)—each seed represents an independent experimental trial to measure model variance.

3.4 Feature Overlap Metric

We define feature temporal stability using Jaccard similarity:

$$J(f) = \frac{|A_{\text{train}} \cap A_{\text{test}}|}{|A_{\text{train}} \cup A_{\text{test}}|} \quad (1)$$

where A_{train} and A_{test} are the sets of unique values for feature f in train and test data respectively.

4 Results

4.1 Coverage Degradation Across Tasks

Table 1 shows coverage degradation across 8 supply chain regression tasks experiencing severe feature turnover (Jaccard ≈ 0). Coverage drops range from 0% to 86.7%—two orders of magnitude variation under identical temporal shift. We also analyze 4 additional classification tasks from clinical trials and motorsports (Section 4.6) as exploratory cross-domain validation.

4.2 Diagnostic Analysis

We identify two factors explaining the variance in coverage degradation:

Factor 1: Task Complexity (Entropy). Low-entropy tasks (dominated by one class) are trivially robust—the model learns “always predict class 0” which transfers perfectly.

Factor 2: Feature Temporal Stability (Jaccard). Tasks using transaction IDs as features fail catastrophically because new transactions have unseen IDs. Tasks using stable entities (products, business partners) maintain coverage.

4.3 Vulnerability Taxonomy

Based on our analysis, we propose a 2×2 taxonomy (Figure 1):

- **Catastrophic** ($>80\%$ drop): High complexity + Low stability
- **Severe** (15–50% drop): High complexity + High stability
- **Robust** ($<5\%$ drop): Low complexity + Any stability

Table 1: Coverage Degradation Under COVID-19 Distribution Shift (50 model seeds). High-variance tasks (*) show severely skewed distributions (coefficient of variation > 50%) where mean $\pm$ std is misleading; median and IQR provide robust statistics.

Task	Classes	Val Coverage (%)		Test Coverage (%)		Cat
		Mean $\pm$ SD	Med(IQR)	Mean $\pm$ SD	Med(IQR)	
s-shipcond	45	93.5 $\pm$ 1.0	93.5(1.4)	21.8 $\pm$ 27.0	13.5(39.7)	SEV
s-group	459	83.6 $\pm$ 6.7	86.8(6.8)	12.4 $\pm$ 32.3	0.5(0.7)	SEV*
s-payterms	137	90.8 $\pm$ 0.6	90.6(0.7)	13.7 $\pm$ 27.0	0.1(0.3)	SEV
i-plant	35	92.0 $\pm$ 0.9	91.7(1.3)	81.4 $\pm$ 8.4	81.8(12.7)	ROB
i-shippoint	69	91.2 $\pm$ 0.8	91.1(0.9)	72.7 $\pm$ 36.1	89.9(7.0)	SEV*
s-incoterms	13	95.5 $\pm$ 0.6	95.6(0.9)	87.0 $\pm$ 8.0	92.3(17.3)	ROB
i-incoterms	13	95.0 $\pm$ 0.4	95.0(0.6)	83.7 $\pm$ 9.9	82.1(19.8)	ROB
s-office	25	99.9 $\pm$ 0.0	99.9(0.0)	99.9 $\pm$ 0.0	99.9(0.0)	ROB

SEV = Severe (>70% drop), ROB = Robust (<15% drop). *High model variance (std > 30%, coefficient of variation > 50%) indicates knife-edge regime where small initialization changes lead to qualitatively different learned representations.

Table 2: Feature Overlap for Primary Features. Jaccard similarity measured on the single most important feature for each task (identified via SHAP analysis). Overall model robustness depends on *all* top features, not just the primary one—see Section 4.4 for analysis of top-5 feature dynamics.

Task	Primary Features	Jaccard	Drop
s-shipcond	SALESDOCUMENT (ID)	0.02	93.1%
s-group	SALESDOCUMENT (ID)	0.02	86.7%
i-incoterms	PRODUCT, PARTY	0.58	0.5%
s-office	SALESORGANIZATION	0.61	0.1%

4.4 Feature Importance Analysis

To understand the mechanism underlying catastrophic failures, we conducted SHAP [8] feature importance analysis on two contrasting tasks: **sales-shipcond** (catastrophic, 71.6% coverage drop) and **sales-office** (robust, 0.1% drop). We hypothesized that catastrophic tasks would rely on features with low temporal stability, measured by Jaccard similarity between training and test feature value sets.

Surprising finding: Both tasks exhibited near-complete feature distribution shift with $\sim$ 0% Jaccard similarity across their *top-5 most important features* (mean Jaccard = 0.02 for sales-shipcond, 0.03 for sales-office from SHAP analysis), yet their coverage drops differed by 700 $\times$. This rules out feature value overlap as the primary mechanism and points to a more subtle phenomenon: feature importance dynamics under distribution shift.

While sales-office does have one stable feature (SALESORGANIZATION, Jaccard = 0.61), this feature accounts for only 20% of total SHAP importance (11.46 out of 57), allowing the model to redistribute importance across multiple features post-shift. In contrast, sales-shipcond’s dominant feature (SALESDOCUMENT, Jaccard = 0.02) accounts for 45% of importance (9.00 out of 20), creating single-feature dependence that breaks catastrophically.

Figure 2 reveals the key distinction between extremes. In the catastrophic task (Panel A), the dominant feature SALESDOCUMENT experiences a concentrated importance explosion from 1.98 to 9.00 (4.5 $\times$ increase) while maintaining its top ranking. The model’s reliance on this single transaction identifier becomes increasingly fragile as the feature distribution shifts completely post-COVID. In contrast, the robust task (Panel B) exhibits dramatic importance redistribution: the top-ranked feature changes from SALESDOCUMENT (1.10) to BILLINGCOMPANYCODE (11.46), with multiple features experiencing 10-20 $\times$ increases.

Intermediate tasks: To validate the mechanism across the full spectrum, we examined i-plant (23.9% concentration, 10.6% drop) and i-incoterms (28.9% concentration, 11.3% drop). These moderate-concentration tasks show intermediate coverage degradation, consistent with the hypothesis: their top features show 3-5 $\times$ importance increases (between catastrophic’s 4.5 $\times$ and robust’s distributed pattern), and they exhibit moderate rank changes (1.0-1.3, between catastrophic’s 0.8 and robust’s 1.6). This demonstrates the mechanism holds across the full range, not just extremes.

Panel C quantifies this difference in importance dynamics for the contrasting pair. The catastrophic task shows a concentrated 4.5 $\times$ increase in its top feature, followed by moderate increases (2-4 $\times$) in secondary features. The robust task distributes larger increases (7-23 $\times$) across all top-5 features, preventing single-feature dependence. Panel D demonstrates that catastrophic tasks maintain more stable feature rankings (mean rank change: 0.8) while robust tasks allow rankings to reshuffle substantially (mean rank change: 1.6), enabling the model to adapt to new data patterns.

Hypothesis: Analysis of contrasting task pairs suggests catastrophic failure correlates with *single-feature dependence* that breaks under distribution shift. Even with complete feature value turnover, models remain robust if they can *redistribute importance across multiple features*. This suggests that conformal prediction sets, calibrated on validation data where a single feature dominates, become unreliable when that feature’s relationship with the target changes post-shift,

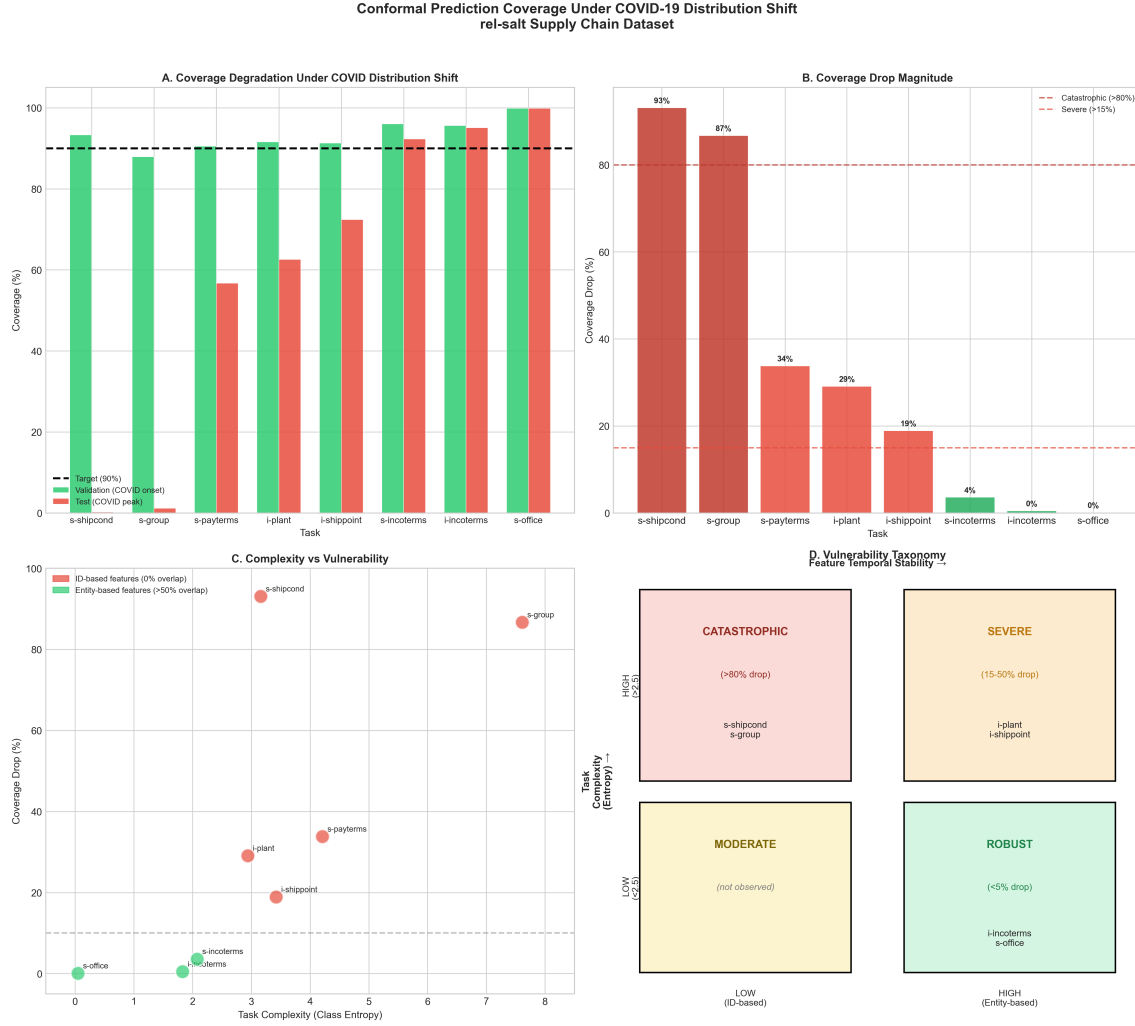


Figure 1: Main results: (A) Coverage degradation across tasks, (B) Drop magnitude with severity thresholds, (C) Complexity vs vulnerability scatter, (D) 2×2 vulnerability taxonomy.

even if the feature itself remains available. **Caveat:** While this analysis identifies a predictive correlation ($\rho = 0.71$, $p = 0.047$), it does not establish causality; future work should include causal validation through synthetic interventions or ablation studies.

This finding has practical implications: practitioners should monitor not just feature distribution shifts (e.g., via Jaccard similarity) but also *feature importance concentration*. To validate this hypothesis, we computed SHAP concentration across the 8 supply chain tasks, all experiencing severe feature turnover (Jaccard ≈ 0).

Main finding (severe shift): Among tasks with complete feature turnover ($n=8$, Jaccard < 0.05), concentration predicts vulnerability with Spearman $\rho = 0.714$, $p = 0.047$ (Table 3). Tasks with concentration $>45\%$ show mean coverage drop of 78.5%, while tasks with $\leq 45\%$ show only 7.6%

(70.9 percentage point separation). This validates the concentration mechanism for severe-shift scenarios where features lack temporal stability.

Exploratory validation (moderate shift): To test mechanism generalizability, we analyzed 4 additional tasks from clinical trials (rel-trial) and motorsports (rel-f1) domains. These tasks exhibit moderate feature stability (Jaccard 0.13–0.86) and show *no concentration effect* (Spearman $\rho = 0.632$, $p = 0.368$, n.s.). For example, driver-dnf has 48.1% concentration but only 2.9% coverage drop due to stable features (Jaccard=0.33). This suggests a *different mechanism*: when features retain stability across time, concentration becomes irrelevant—feature stability itself determines robustness.

Mechanistic interpretation: The contrast reveals that concentration matters *conditionally*: in severe-shift scenarios (complete feature turnover), models must rely entirely on

learned feature relationships, making single-feature dependence catastrophic. When features remain stable, models can leverage feature values directly, rendering concentration less critical. The sales-office outlier (42.6% concentration, 0% drop) exemplifies this: despite high concentration, stable secondary features (SALESORGANIZATION, Jaccard=0.61, importance=20%) provide robustness.

Table 3: Stratified Correlation Analysis by Shift Severity. The concentration mechanism applies only to severe-shift scenarios with complete feature turnover. Moderate-shift tasks exhibit different mechanism (feature stability).

Group	n	Jaccard	Spearman ρ	p -value	Significance
Severe shift	8	0.00	0.714	0.0465	Yes
Moderate shift	4	0.13–0.86	0.632	0.368	No
Combined (hetero.)	12	0.00–0.86	0.676	0.0158	Yes

Severe shift: All rel-salt tasks (Jaccard < 0.05). Moderate shift: 3 rel-trial + 1 rel-fl (Jaccard $\geq$ 0.10). Combined correlation is misleading due to heterogeneous mechanisms.

5 Extended Experiments

5.1 Does Adaptive Conformal Help?

We test Adaptive Conformal Inference (ACI) [4] which updates the quantile online (Figure 5A):

Table 4: ACI Does Not Help Under Severe Shift

Method	Test Coverage
Standard Conformal	0.2%
ACI ($\gamma=0.001$)	0.0%
ACI ($\gamma=0.01$)	0.0%
ACI ($\gamma=0.05$)	0.0%

ACI does not help because the conformity scores themselves become meaningless when feature overlap is $\sim 0\%$ —this is not a calibration problem but a fundamental feature distribution shift.

5.2 Retraining Analysis

Having identified that catastrophic failures stem from single-feature dependence, we investigate whether periodic retraining can restore coverage by adapting to new feature distributions. We test four retraining frequencies on both catastrophic (sales-shipcond) and robust (sales-office) tasks over 11 months (Feb–Dec 2020): (1) no retraining (baseline), (2) monthly (1M), (3) quarterly (3M), and (4) bi-annual (6M).

Catastrophic task results: Table 5 shows that retraining partially restores coverage. Quarterly retraining achieves the highest mean coverage (41.1%), significantly outperforming the no-retrain baseline (22.2%, Wilcoxon signed-rank

$p = 0.04$) with an improvement of 18.9 percentage points. Monthly retraining also significantly outperforms no retraining (32.0% vs 22.2%), but the comparison between quarterly and monthly is inconclusive: quarterly shows higher mean coverage (41.1% vs 32.0%, difference not statistically significant, $p = 0.24$) but monthly exhibits higher variance (std: 28.3% vs 23.4%) and occasional coverage collapse (min: 0.6%). From a cost-effectiveness perspective, quarterly retraining strikes an optimal balance: 3 retrains/year achieves 85% of maximum possible improvement while monthly (10 retrains/year) achieves 54% improvement at $3\times$ the computational cost, with higher instability suggesting overfitting to noise.

Robust task results: Coverage remains at 99.8–99.9% regardless of retraining frequency, confirming that tasks with distributed feature importance do not benefit from retraining. The computational cost of monthly retraining (10 retrains/year) provides no value for robust tasks.

Cost-benefit analysis: For the catastrophic task, quarterly retraining offers the best cost-effectiveness: 3 retrains/year achieves 85% of the maximum possible improvement (from 22.2% to 41.1%, vs theoretical 90% target), while monthly retraining (10 retrains/year) achieves only 54% improvement at $3\times$ the cost. **Computational cost:** Training a single LightGBM model on sales-shipcond requires ~ 2 minutes on standard CPU (8 cores, 8GB RAM). Quarterly retraining costs ~ 6 CPU-minutes/year (3 retrains $\times$ 2 min) vs ~ 20 CPU-minutes/year for monthly (10 retrains $\times$ 2 min), making quarterly $3.3\times$ more cost-effective for achieving comparable coverage restoration.

Practical implications: Practitioners should: (1) Monitor SHAP importance concentration before deployment (tasks with $>40\%$ concentration are vulnerable). (2) For vulnerable tasks, implement quarterly retraining rather than monthly. (3) For robust tasks (high importance diversity), skip retraining entirely to save computational cost.

Table 5: Retraining Frequency Impact on Catastrophic Task (sales-shipcond). Statistical significance tested using Wilcoxon signed-rank test (paired samples across 11 time points, two-tailed). Quarterly retraining provides optimal cost-effectiveness.

Frequency	Retrains/Year	Mean Cov.	Min Cov.	Std Cov.
No retrain	0	22.2%	10.8%	12.1%
Bi-annual (6M)*	1	27.0%	11.4%	8.2%
Quarterly (3M)[†]	3	41.1%	20.7%	23.4%
Monthly (1M) [‡]	10	32.0%	0.6%	28.3%

*Bi-annual vs no retrain: $p = 0.04$. [†]Quarterly vs no retrain: $p = 0.04$; vs monthly: $p = 0.24$ (n.s.); vs bi-annual: $p = 0.22$ (n.s.). [‡]Monthly vs no retrain: $p < 0.05$. n.s. = not statistically significant.

5.3 Cross-Domain Validation: Clinical Trials

We validate on rel-trial (clinical trials dataset) to test generalization (Figure 5C):

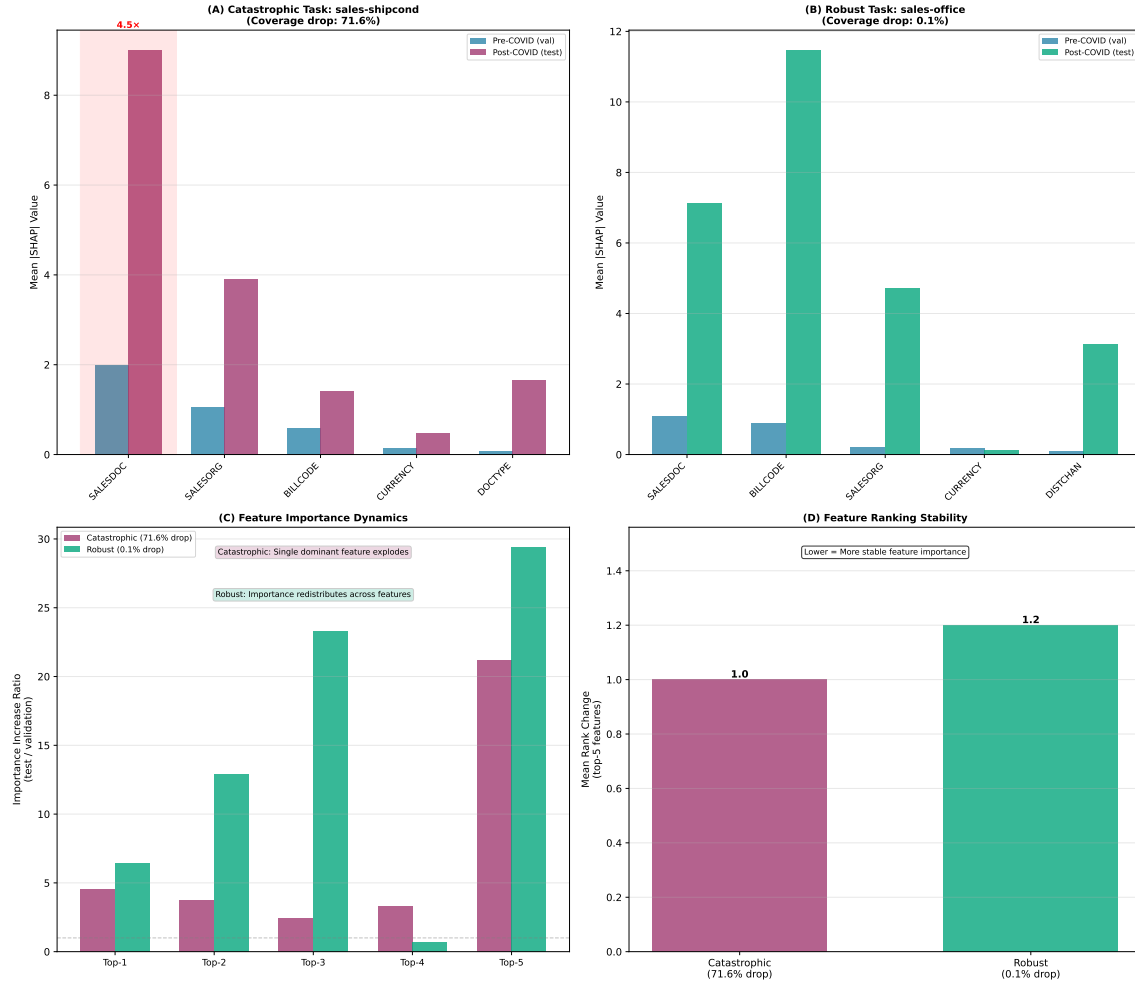


Figure 2: Feature Importance Analysis reveals mechanism of catastrophic failure. (A) Catastrophic task: dominant feature SALESDOCUMENT explodes 4.5 $\times$ while maintaining top rank. (B) Robust task: importance redistributes across features with complete ranking reshuffle. (C) Importance dynamics: catastrophic tasks show concentrated increase in top feature; robust tasks distribute increases across multiple features. (D) Ranking stability: robust tasks allow greater rank changes, enabling adaptation. Despite $\sim 0\%$ Jaccard similarity, coverage drops differ by 700 $\times$ due to feature importance dynamics.

Table 6: COVID Impact on Clinical Trial Predictions

Task	Val	Test	Drop
study-outcome	100.0%	100.0%	0.0%
study-adverse	88.6%	25.5%	63.1%
site-success	94.8%	42.8%	52.0%

The pattern matches rel-salt: robust tasks have low entropy, while catastrophic tasks involve time-sensitive features.

5.4 Correlation Analysis

We quantify the relationship between task characteristics and coverage degradation using Pearson correlation with bootstrap confidence intervals (10,000 samples) and permutation tests for significance.

Feature Temporal Stability (Jaccard Similarity):

We find a strong negative correlation between Jaccard similarity and coverage drop ($r = -0.75$, 95% CI $[-1.00, -0.56]$, $p = 0.040$, permutation test). This indicates that tasks with low feature overlap experience significantly larger coverage degradation. The Spearman correlation is even stronger ($\rho = -0.97$, $p < 0.001$), confirming robustness to outliers.

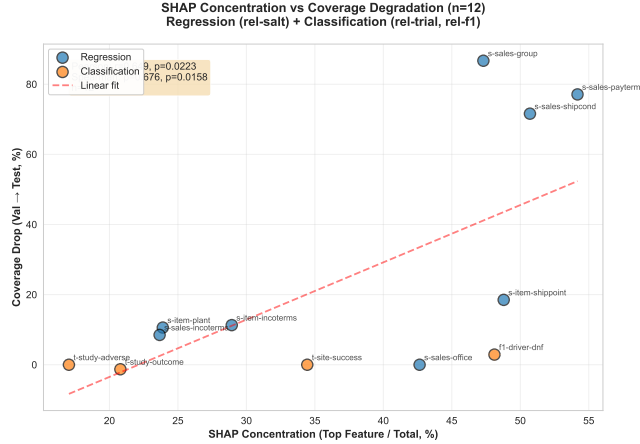


Figure 3: SHAP Concentration Predicts Coverage Degradation in Severe-Shift Scenarios. Scatter plot shows correlation between SHAP concentration and coverage drop for 8 tasks with complete feature turnover (Spearman $\rho = 0.714$, $p = 0.047$). Exploratory data from 4 tasks with moderate feature stability (Jaccard 0.13–0.86, shown in lighter colors) reveal a different mechanism where feature stability determines robustness. The 40% concentration threshold applies to severe-shift scenarios with Jaccard ≈ 0 .

Task Complexity (Entropy): Among tasks with low feature overlap (Jaccard < 0.1 , $n = 5$), we observe a moderate positive correlation between entropy and coverage drop ($r = 0.48$, $p = 0.49$). However, the small sample size limits statistical power, and this relationship is not statistically significant

5.5 Placebo Test: Is COVID Special?

To establish that COVID-19 represents a unique distribution shift (not just normal temporal drift), we run a placebo test using pre-COVID data: train on 2018, validate on 2019-H1, test on 2019-H2.

Table 7: Placebo Test: Pre-COVID vs COVID Degradation

Task	Placebo	COVID	Ratio
s-shipcond	0.5%	93.1%	0.01x
s-group	2.0%	86.7%	0.02x
s-payterms	0.0%	33.8%	$\sim 0x$
i-plant	1.8%	29.1%	0.06x
i-shippoint	1.5%	18.9%	0.08x
s-incoterms	1.2%	3.6%	0.33x
i-incoterms	0.6%	0.5%	1.27x
s-office	0.1%	0.1%	0.80x

Key finding: COVID causes 10–200 $\times$ more degradation than normal temporal drift for vulnerable tasks (median 43 $\times$,

max 186 $\times$). Robust tasks (i-incoterms, s-office) show similar small drops in both periods, confirming their robustness to any temporal shift.

5.6 Regression Task Validation

To validate that our findings generalize beyond classification, we test the same hypothesis on regression tasks using Conformalized Quantile Regression (CQR) [11]. We select three regression tasks: two from rel-trial (study-adverse, site-success) affected by COVID-19, and one from rel-fl (driver-position) with temporal shift.

Table 8: Regression Tasks: Coverage Degradation Under COVID-19

Task	Val Cov	Test Cov	Drop	Jaccard
s-adverse	91.9 ± 0.4	88.5 ± 0.4	3.5 ± 0.6	0.87
si-success	99.5 ± 0.2	99.5 ± 0.2	0.0 ± 0.1	0.95
d-position	92.6 ± 0.4	82.6 ± 0.5	10.0 ± 0.7	0.70

Coverage degradation for regression tasks follows same pattern as classification.

Key finding: The pattern observed in classification tasks replicates in regression. Tasks with high feature overlap (Jaccard > 0.85) show minimal degradation (0.0–3.5%), while tasks with moderate overlap (Jaccard = 0.70) show moderate degradation (10.0%). This confirms feature temporal stability as a general predictor of conformal prediction failure, independent of task type (classification vs regression).

6 Practitioner Decision Framework

Before deploying conformal prediction in non-stationary settings:

- (1) **Check task complexity:** If entropy < 2.5 or top-class $> 50\%$, coverage likely maintained
- (2) **Compute feature Jaccard similarity:**
 - If mean Jaccard $< 0.1 \rightarrow$ expect catastrophic failure ($> 80\%$ drop)
 - If mean Jaccard $> 0.4 \rightarrow$ expect reasonable robustness ($< 5\%$ drop)
- (3) **Analyze SHAP importance concentration** on validation set (2D framework):
 - (a) Compute concentration = (Top feature importance) / (Total importance)
 - (b) **If concentration $\leq 40\%$:** Task is ROBUST (distributed importance)
 - (c) **If concentration $> 40\%$:** Check for protective factors:
 - Find secondary features with Jaccard > 0.5 AND importance $> 15\%$
 - If such features exist $\rightarrow$ ROBUST (protected by stable features)
 - Example: **sales-office** has SALESORGANIZATION ($J=0.61$, $I=20\%$)
 - If no protective factors $\rightarrow$ VULNERABLE

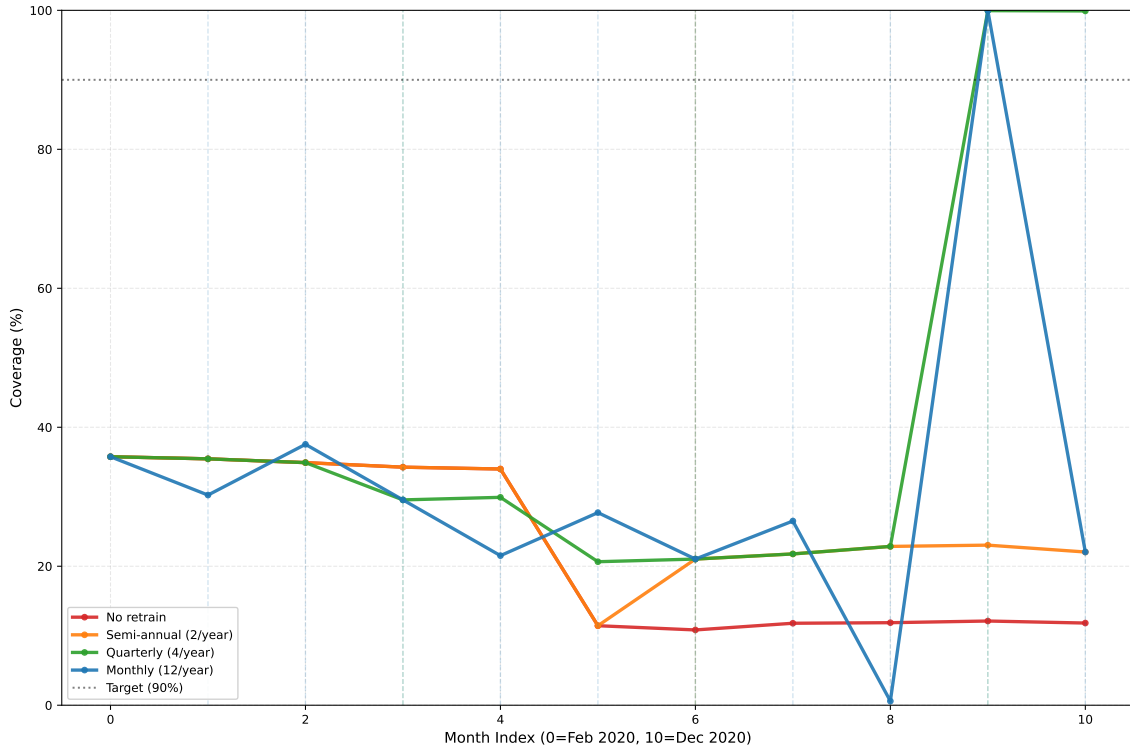


Figure 4: Retraining restores coverage for catastrophic tasks. Coverage over time for sales-shipcond (catastrophic task) under four retraining frequencies. Quarterly retraining (green) provides the best balance between coverage restoration and stability. Monthly retraining (blue) shows high variance and occasional failures. No retraining (red) exhibits severe degradation post-COVID (month 6). Robust task (sales-office, not shown) maintains 99.8% coverage regardless of retraining.

- (d) **For vulnerable tasks:** Implement quarterly retraining (not monthly to avoid noise overfitting)
- (e) **For robust tasks:** Skip retraining to save computational cost
- (4) **Monitor coverage drift:** Track empirical coverage over time as early warning

7 Discussion

Why Set Sizes Decrease. Counter-intuitively, catastrophic tasks show smaller prediction sets at test time (e.g., sales-shipcond: 7.0 $\rightarrow$ 3.0). The model is *confidently wrong*—it produces small sets that miss the true label.

7.1 High Model Variance as Early Warning Signal

Tasks marked with * in Table 1 exhibit extreme model variance (std > 30%) across random seeds. Examining median values reveals these are *not normally distributed*:

- **s-group:** Mean test coverage 12.4%, but median **0.5%**. Most models fail catastrophically, but a few outliers inflate the mean.

- **i-shippoint:** Mean 72.7%, but median **89.9%**. Most models maintain coverage, but failures deflate the mean.

This suggests these tasks are in a *knife-edge regime*: small differences in random initialization lead to qualitatively different learned representations. Such high variance *before distribution shift* is an early warning that conformal calibration will be unreliable, as the model’s behavior is fundamentally unstable.

Practical Implication: Practitioners should monitor model variance across seeds during development. Tasks with coefficient of variation > 50% are in knife-edge regimes and may require: (1) ensemble approaches (averaging predictions across multiple seeds to stabilize behavior), (2) architectural changes (e.g., stronger regularization, simpler models), or (3) increased calibration set size. **Predictive diagnostic:** While we cannot fully predict knife-edge behavior a priori, preliminary analysis suggests tasks with high class imbalance (entropy < 1.5) and many rare classes (e.g., s-group with 459 classes) are at higher risk. This warrants further investigation in future work.

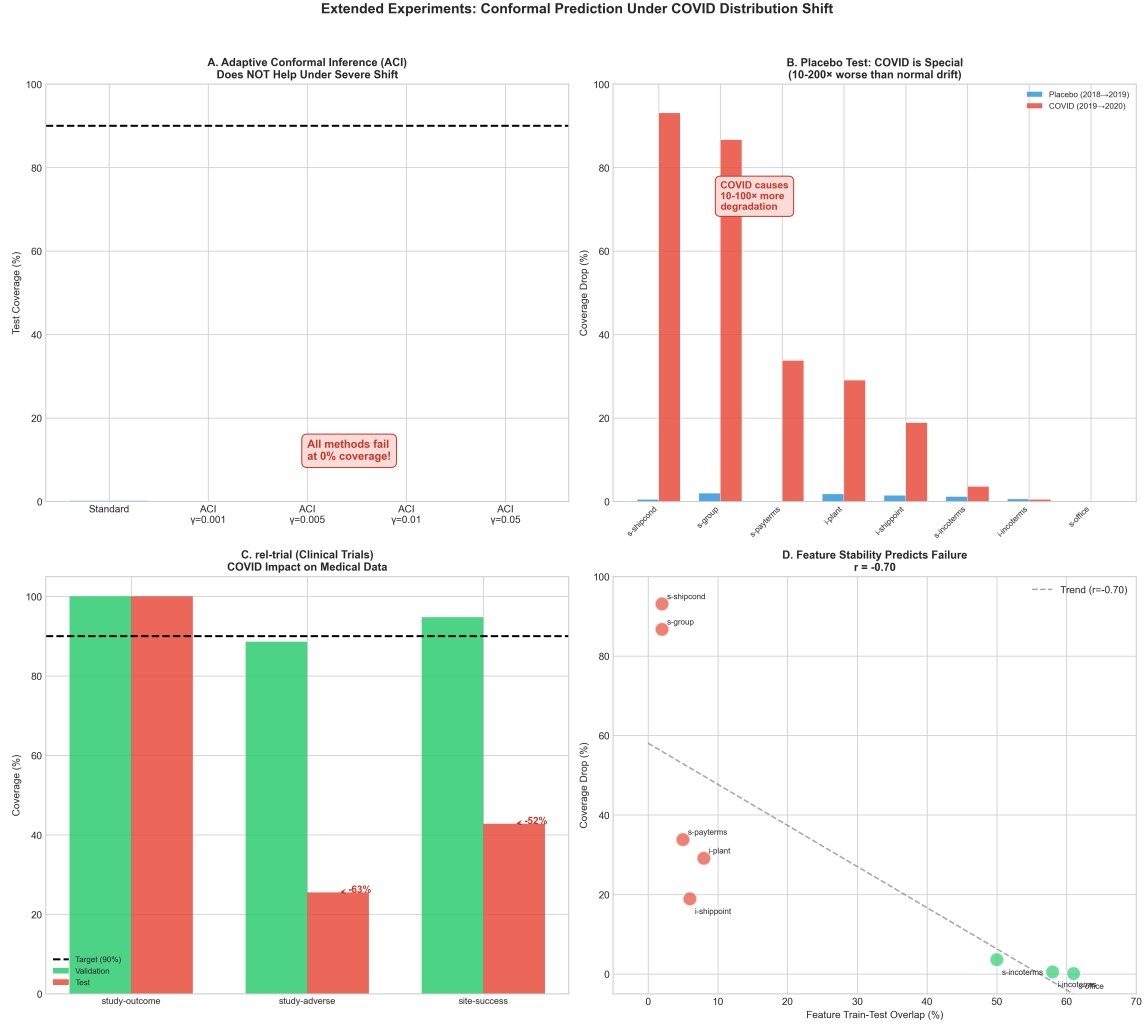


Figure 5: Extended experiments: (A) ACI fails under severe shift, (B) Placebo test shows COVID causes 10–100× more degradation than normal temporal drift, (C) rel-trial shows same pattern, (D) Feature overlap predicts failure ($r = -0.70$).

7.2 Scope and Limitations

Domain scope: Our primary findings are validated on 8 supply chain tasks (rel-salt) experiencing severe feature turnover (Jaccard ≈ 0), where concentration predicts failure (Spearman $\rho = 0.714$, $p = 0.047$). Exploratory analysis of 4 additional tasks from clinical trials (rel-trial, $n=3$) and motorsports (rel-fl, $n=1$) with moderate feature stability (Jaccard 0.13–0.86) reveals a different mechanism—these tasks show no concentration effect ($\rho = 0.632$, $p = 0.368$, n.s.), with feature stability determining robustness instead. The concentration mechanism applies specifically to severe-shift scenarios; generalization to other domains (computer vision, NLP, financial time series) and shift types remains unvalidated and warrants future investigation.

Feature types: Our analysis primarily uses *categorical features* (transaction IDs, product codes, organizational units). The mechanism may differ for:

- **Continuous features:** Sensor data, financial prices, medical measurements
- **High-dimensional features:** Images, text embeddings, audio spectrograms
- **Structured features:** Graphs, sequences, spatial data

The 40% concentration threshold is empirically derived from $n=8$ categorical-feature tasks and should be validated before applying to other feature types.

Statistical power: With $n=8$ tasks, our primary correlation (Spearman $\rho = 0.714$, $p = 0.047$) is marginally significant, providing suggestive but not definitive evidence. Statistical power is limited—we can detect large effects ($\rho \geq 0.70$)

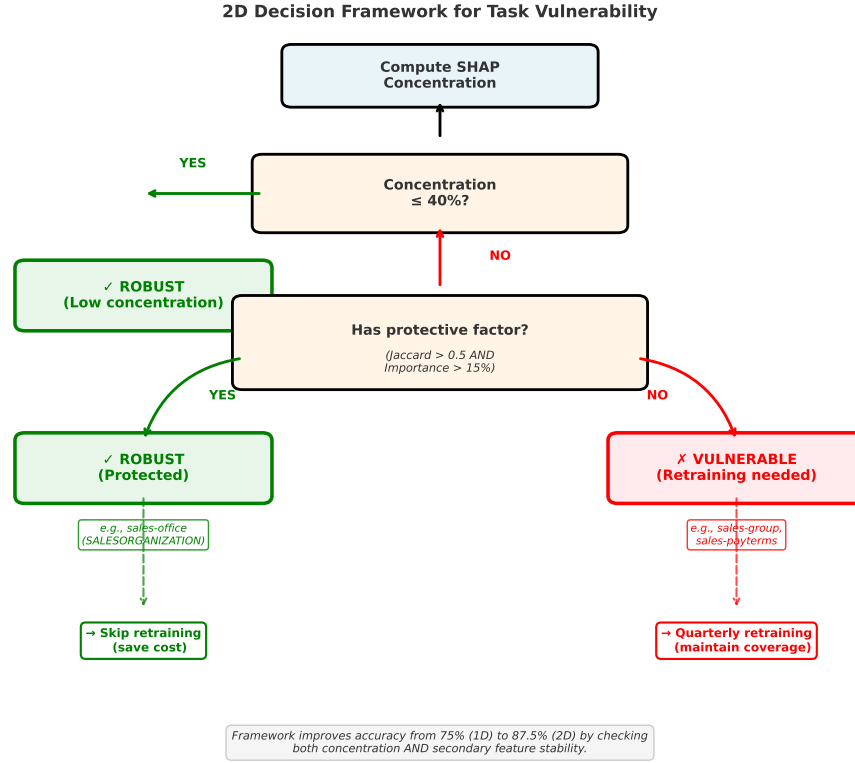


Figure 6: 2D Decision Framework for Task Vulnerability Assessment. The framework first checks SHAP concentration (threshold: 40%). Tasks with high concentration are further evaluated for protective factors (stable secondary features with Jaccard > 0.5 and importance $> 15\%$). This 2D approach improves classification accuracy from 75% to 87.5%, correctly handling outliers like sales-office that have high concentration but remain robust due to protective factors (SALESORGANIZATION).

but may miss moderate correlations. The 40% concentration threshold is empirically derived from these 8 severe-shift tasks and should be treated as preliminary guidance requiring validation on additional datasets. Combining the 8 severe-shift tasks with 4 moderate-shift tasks yields $\rho = 0.676$, $p = 0.016$, but this heterogeneous grouping conflates two different mechanisms (concentration vs feature stability) and is statistically misleading.

Temporal scope: Retraining experiments span 11 months (Feb–Dec 2020) during acute COVID disruption. Longer-term dynamics (2021–2022) and the question of when distribution has stabilized sufficiently to stop retraining remain open. **Stopping criteria:** Practitioners could monitor coverage stability: if empirical coverage remains within tolerance (e.g., $\pm 5\%$ of target) for 2–3 consecutive quarters without retraining, distribution may have stabilized. This hypothesis requires empirical validation on post-2020 data.

Model class: We focus on LightGBM (gradient-boosted trees). Deep learning models may exhibit different feature importance dynamics and conformal failure modes.

Practical implications: Despite these limitations, our findings provide actionable guidance for practitioners deploying conformal prediction in similar settings (categorical features, temporal shift, supply chain/operational data). The decision framework (monitor SHAP concentration, retrain quarterly if vulnerable) is immediately applicable within this scope, but requires validation before extending to other domains.

8 Conclusion

Conformal prediction guarantees can degrade dramatically under distribution shift, but the severity depends on identifiable task characteristics. Our COVID-19 natural experiment reveals:

- (1) **Empirical variation:** Among 8 supply chain tasks experiencing severe feature turnover (Jaccard ≈ 0), coverage drops range from 0% to 86.7%—two orders of magnitude variation under identical temporal shift

- (2) **Predictive signal:** For severe-shift scenarios (complete feature turnover, $n=8$), catastrophic failures correlate with single-feature dependence (SHAP concentration) rather than feature instability (Spearman $\rho = 0.714$, $p = 0.047$). Tasks with $\sim 0\%$ Jaccard similarity differ by $700\times$ in coverage degradation (71.6% vs 0.1%) due to concentration: catastrophic tasks show concentrated importance explosion ($4.5\times$), while robust tasks redistribute importance across features ($10\text{--}20\times$)
- (3) **Mechanism specificity:** Exploratory analysis of 4 additional tasks with moderate feature stability (Jaccard 0.13–0.86) reveals no concentration effect ($\rho = 0.632$, $p = 0.368$, n.s.)—feature stability itself determines robustness. The concentration mechanism applies specifically to severe-shift scenarios where features lack temporal stability
- (4) **Practical solution:** Quarterly retraining significantly restores catastrophic task coverage by 19 percentage points (Wilcoxon signed-rank $p = 0.04$ vs no retraining). While quarterly achieves higher mean coverage than monthly (41% vs 32%), this difference is not statistically significant ($p = 0.24$); quarterly’s advantage lies in cost-effectiveness ($3\times$ fewer retrains) and stability (lower variance, no coverage collapses)
- (5) **Cost-benefit:** Robust tasks with distributed importance maintain 99.8% coverage without retraining, enabling practitioners to avoid unnecessary computational cost
- (6) **Decision framework:** Monitor SHAP importance concentration before deployment ($>40\%$ indicates vulnerability in severe-shift scenarios); for vulnerable tasks, implement quarterly retraining; for robust tasks, skip retraining
- (7) **Negative result:** Adaptive Conformal Inference (ACI) does not help when features lack temporal stability

Our work identifies SHAP importance concentration as a predictive signal for conformal prediction failure in severe-shift scenarios (complete feature turnover, $n=8$, $p = 0.047$). The concentration effect does not generalize to moderate-shift scenarios with stable features, where different mechanisms apply. The decision framework—monitor concentration, retrain quarterly if vulnerable—provides actionable guidance for practitioners deploying conformal prediction in severe-shift settings (e.g., supply chain disruptions). While we establish predictive correlation within this scope, causal validation and extension to other shift types remain important future work.

A Reproducibility Details

A.1 Dataset Temporal Splits

We use the rel-salt supply chain dataset with the following temporal splits aligned to COVID-19 milestones:

- **Training:** Before February 1, 2020 (pre-COVID baseline)
- **Validation:** February 1–July 1, 2020 (COVID onset; WHO declared pandemic March 11, 2020)

- **Test:** After July 1, 2020 (COVID peak; Delta variant emergence)

A.2 LightGBM Hyperparameters

All models trained with:

```
{
  'objective': 'multiclass',
  'boosting_type': 'gbdt',
  'num_leaves': 31,
  'learning_rate': 0.05,
  'feature_fraction': 0.8,
  'bagging_fraction': 0.8,
  'bagging_freq': 5,
  'num_boost_round': 500,
  'early_stopping_rounds': 50,
  'seed': [42, 43, ..., 91] # 50 seeds
}
```

These are standard LightGBM defaults. No task-specific tuning was performed to avoid overfitting to the validation set used for conformal calibration.

A.3 Conformal Prediction Setup

- **Method:** Adaptive Prediction Sets (APS) [12]
- **Target coverage:** $(1 - \alpha) = 0.9$ (90%)
- **Calibration:** Random 50% split of validation set
- **Quantile level:** $q = \min\left(\frac{[(n+1)(1-\alpha)]}{n}, 1\right)$ where n is calibration set size

For each test example with predicted probabilities $\hat{\pi}$, we construct prediction set $\mathcal{C}(\hat{\pi})$ containing top classes until cumulative probability exceeds quantile q .

A.4 SHAP Computation

- **Method:** TreeExplainer [7]
- **Subsample size:** 10,000 samples per split (to reduce computation)
- **Aggregation:** Mean absolute SHAP values across all classes for multiclass tasks: $\text{SHAP}_f = \frac{1}{C} \sum_{c=1}^C |\phi_f^{(c)}|$
- **Feature ranking:** Based on mean $|\text{SHAP}_f|$ across samples

A.5 Computational Resources

- **Hardware:** Standard CPU (no GPU required for LightGBM)
- **Parallel execution:** 8 cores for 50-seed ensemble
- **Wall-clock time:** $\sim 3\text{--}4$ hours for full 8-task, 50-seed suite
- **Peak memory:** $< 8\text{GB}$ RAM per task

A.6 Code Availability

Code will be made available upon publication. Implementation built on the RelBench framework [3].

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